Safari Research Proposal

SD5008 Research and Analysis for Design | ISD

Group Name

Safari

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Research Project Title

Attention Allocation and Navigation Information Design: for Newcomers to Hong Kong

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Research Background

Walking navigation involves interpreting directions while monitoring the surroundings and deciding what to do next. For newcomers encountering unfamiliar routes in Hong Kong, the relationship between digital guidance and environmental cues provides a useful setting for studying these demands. We will investigate when navigation information supports action and when obtaining, interpreting or responding to it creates additional interaction burden. We will not assume that all newcomers experience excessive burden.

Multiple resource theory explains how concurrent tasks can interfere when they draw on overlapping processing resources (Wickens, 2008). Oulasvirta et al. (2005) found fragmented attention during mobile web-search tasks in urban settings. Although this was not a navigation study, it supports examining interaction as a sequence of situated episodes. Wobbrock (2019) further shows how environmental and bodily conditions can constrain technology use. These perspectives will guide interpretation rather than serve as mechanisms that this small study claims to prove.

Recent navigation studies suggest that reducing visual information or changing modality does not automatically improve every outcome. Bartling et al. (2026, published online in 2025) found increased self-reported cognitive workload under heavy virtual traffic, but only partial support for maps that masked task-irrelevant points of interest. Nagata et al. (2026, published online in 2025) compared voice navigation with paper maps among adults aged 50–69: voice guidance improved relaxation, while fatigue and navigation performance did not significantly improve. Neither study directly establishes the experience of newcomers in Hong Kong.

Building on this work, we will develop an empirically grounded classification connecting the role of navigation information with its contextual interaction demands. The proposed contribution is a design framework linking information needs to presentation timing and format. The literature review will further assess this contribution; a universal research gap is not assumed.

Research Objectives

- Identify navigation information needs and attention-related interaction episodes during unfamiliar walking journeys.
- Classify information by its task role and the effort involved in finding, interpreting, remembering, confirming and acting on it.
- Examine how environmental conditions, route familiarity and presentation choices relate to experienced burden and coping strategies.
- Translate the findings into a preliminary design framework and examine its clarity and usefulness through participant feedback on navigation scenarios.

Research Questions

Main question: How can navigation information be classified by its associated interaction burdens to inform interaction design methods for improving when and how it is presented and delivered?

- What information do newcomers need at different stages of a walking navigation task, and how do they coordinate attention between the navigation interface and their surroundings?
- Under what conditions do navigation information and its presentation create additional effort, uncertainty or interruption, and how do users respond?
- How can an evidence-based classification guide decisions about information timing, detail and delivery format while preserving understanding and confidence?

Data Collection Methods

We propose an exploratory qualitative study supported by descriptive behavioural records and brief self-ratings. The unit of analysis will be an information-interaction episode: a moment when a participant seeks, receives, interprets, confirms or acts on navigation information. Interaction burden is provisionally defined as the effort and interruption involved in these activities under particular task and environmental conditions. Necessary confirmation will not automatically be classified as excessive burden.

Background review and pilot

We will map relevant attention theories, pedestrian navigation studies and existing navigation interfaces. We will then pilot the interview guide, route and observation record with two eligible volunteers. Pilot data will be excluded from the main analysis if the procedure changes materially. Piloting and participant research will follow the course's applicable ethics process.

Interview and observed walking journey

Each main session will last approximately 50–75 minutes: a 10–15-minute interview, a 20–30-minute walking task and a 20–30-minute retrospective discussion. The interview will establish navigation habits, time living in Hong Kong, language familiarity, route familiarity and recent difficult navigation experiences.

Participants will use their usual smartphone navigation tools to locate a public destination from a transport exit. One of two pre-scouted routes of comparable length and task structure will be selected according to unfamiliarity. Both routes will include ordinary decision points and a destination entrance. The scope includes initial orientation, walking, stopping to check, entrance finding and recovery from confusion; vehicle travel and complex multi-stop planning are excluded.

A researcher will record timestamps, visible phone interactions, apparent shifts between phone and surroundings, pauses, repeated checks, requests for help and route corrections. These are observational indicators, not precise eye-tracking measures. Researchers will not require continuous think-aloud speech while walking. Questions will be deferred to safe stopping points or the retrospective discussion; sessions will be stopped or rescheduled if conditions are unsafe.

During the retrospective discussion, participants will explain selected episodes using the researcher's timeline and, if they choose, relevant screens with personal content hidden. For key episodes, they will rate perceived mental effort and uncertainty on separate 1–7 scales with verbal anchors. These exploratory ratings will support discussion rather than act as validated measures of cognitive load or fatigue. Interview audio will be recorded only with separate consent; notes will be used if recording is declined.

Formative scenario review

After initial analysis, we will invite 4–6 main-study participants to a separate 20–30-minute review. Two or three anonymised scenarios will illustrate observed problems and alternative timing or presentation choices. We will ask participants to interpret the guidance, explain their next action and discuss confidence, effort and missing information. Presentation order will be alternated. This review will refine the framework; it will not establish real-world performance benefits or modality superiority.

Data analysis

We will combine theory-informed coding with categories emerging from interviews and observation. Initial information roles will include orientation, action guidance, status confirmation and error recovery. Candidate demands include searching, interpretation, map-to-environment matching, remembering, switching attention and operating the device. These categories are provisional and may overlap or change.

For each episode, we will link the task, information role, environmental conditions, observed behaviour, participant explanation, reported effort and outcome. Two team members will independently code a subset, discuss differences and revise a shared codebook before coding the remaining material. We will examine contrasting cases, including occasions when more information or an extra check reduces uncertainty. Counts and ratings will be descriptive. We will maintain a traceable link from evidence to category to design recommendation, without claiming causal effects or statistical generalisation.

Who the Participants Are

We propose recruiting 8–12 adults aged 18 or above who moved to Hong Kong within the previous six months, use smartphone navigation and can undertake the selected walking task comfortably. The six-month threshold is a practical recruitment definition, not an established cognitive boundary. Participants must be unfamiliar with the selected route and able to discuss their experience in a language supported by the research team, provisionally English or Mandarin. We will record route and tool familiarity separately from newcomer status.

Purposive convenience sampling will seek variation in navigation habits and self-reported direction confidence. The sample is intended for detailed exploration, not population estimates or a claim of thematic saturation. Recruitment through university networks may overrepresent students; findings will be reported as applying to the recruited sample. Research team members will not be main-study participants.

How Participants Will Be Recruited

We will circulate an invitation through university student networks and newcomer communities where posting is permitted, and ask volunteers to share it with eligible peers. Interested individuals will contact the team privately and complete a short eligibility screening. Participation will be voluntary and unrelated to grades or access to services. No payment is proposed at this stage. Screening will collect only eligibility and scheduling information; contact details will be stored separately from study records.

Whether Participants Will Receive an Information Sheet and How

Yes. Participants will receive a digital information sheet in a supported language before scheduling and a copy at the session. It will explain the purpose, activities, duration, expected walking conditions, recording options, data use, confidentiality, withdrawal process and researcher and supervisor contacts. The team will answer questions and obtain written or electronic informed consent before collecting research data.

Participants may skip questions, decline audio recording, pause or stop without penalty. Optional follow-up participation and permission to use de-identified quotations will be addressed explicitly. Participants may request removal of identifiable study data within seven days of their session; the information sheet will explain that removal may no longer be possible after irreversible anonymisation and aggregation, which will occur only after that withdrawal window. Tutor and contact details will be completed before recruitment.

Whether Information Will Be Kept Confidential and How

Yes. Participants will be assigned codes such as P01. Names, contact details and consent records will be separated from transcripts and episode records. Reports will use de-identified quotations and generalised journey descriptions, excluding home addresses, account details and identifying combinations of personal information. Private messages, payment details and location histories will not be collected. Continuous video or screen recording is not planned.

Data will be stored in access-restricted, institution-approved storage accessible only to the student research team and authorised supervisors. Raw participant data will not be uploaded to public services or external AI tools. Audio will be deleted after transcription and accuracy checks under the approved protocol. The retention period for consent forms, coded data and contact records will be confirmed with the cohort tutor and stated in the information sheet before recruitment. Confidentiality will be maintained within institutional requirements; participation in a public walking task cannot itself be fully concealed.

Anticipated Data Types

- Screening and contextual data: newcomer eligibility, navigation habits, language and route familiarity.
- Qualitative data: consented interview audio, de-identified transcripts, field notes, participant explanations and scenario-review comments.
- Episode records: information roles, presentation conditions, task stage, visible actions, contextual triggers and outcomes.
- Descriptive numeric data: observed event counts, approximate timings, and episode-level effort and uncertainty ratings.
- Analytical outputs: coded episode matrices, journey maps, the evolving classification and evidence-linked design recommendations.

Research Timeline

The following is a proposed six-week work plan, to be aligned with the confirmed course submission dates. Participant recruitment and data collection will begin only after the required tutor and ethics clearance. Any delay will shift the participant phases rather than bypass review.

Period	Activities and outputs
Week 1	Refine literature mapping, scope and research questions; prepare the protocol, information sheet, consent form and route plan for review.
Week 2	After clearance, recruit and pilot; revise the interview guide and observation record.

Period	Activities and outputs
Weeks 3–4	Conduct 8–12 main sessions; transcribe and code iteratively; compare episodes and draft the classification.
Week 5	Conduct 4–6 scenario reviews; refine design recommendations and examine contradictory evidence.
Week 6	Complete synthesis, limitations, research report and presentation; audit consent, quotations and data handling.

Expected Outcome

The study is expected to produce a contextual classification of navigation information and interaction demands, an account of when and why participants experience difficulty or effective support, and a preliminary interaction design framework. The framework will help designers identify the current task, determine the information needed, consider competing environmental demands, and select presentation timing, detail and format while retaining access to confirmation and recovery.

Illustrated scenarios will show how recommendations follow from observed episodes and participant explanations. The intended contribution is a situated, evidence-based method for reasoning about navigation information, rather than a completed navigation product. Any benefits of proposed interactions will require further field evaluation. Small-scale recruitment, observer effects, retrospective recall, differences between navigation applications and the limited routes will constrain the findings; applicability beyond the sampled newcomers and walking contexts remains to be tested.

Key References

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